

Sonnet: Spectral Operator Neural Network for Multivariable Time Series Forecasting

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Abstract

Multivariable time series forecasting methods can integrate information from exogenous variables, leading to significant prediction accuracy gains. The transformer architecture has been widely applied in various time series forecasting models due to its ability to capture long-range sequential dependencies. However, a naïve application of transformers often struggles to effectively model complex relationships among variables over time. To mitigate against this, we propose a novel architecture, termed **Spectral Operator Neural Network (Sonnet)**. Sonnet applies learnable wavelet transformations to the input and incorporates spectral analysis using the Koopman operator. Its predictive skill relies on the **Multivariable Coherence Attention (MVCA)**, an operation that leverages spectral coherence to model variable dependencies. Our empirical analysis shows that Sonnet yields the best performance on 34 out of 47 forecasting tasks with an average mean absolute error (MAE) reduction of 2.2% against the most competitive baseline. We further show that MVCA can remedy the deficiencies of naïve attention in various deep learning models, reducing MAE by 10.7% on average in the most challenging forecasting tasks.

Code & data — <https://github.com/ClaudiaShu/Sonnet>

Appendices — <https://arxiv.org/pdf/2505.15312>

1 Introduction

Multivariable time series (MTS) forecasting methods learn from multiple input variables to predict a single target variable (Hidalgo and Goodman 2013). MTS models are deployed in many real-life applications, such as financial modelling (Niemira and Saaty 2004; Antunes et al. 2018; Taylor and Buizza 2003; Witt and Witt 1995), numerical weather prediction (Fildes and Kourentzes 2011; Young 2018; Dalton and Bekker 2022), and computational epidemiology (Dugas et al. 2013; Shaman and Karspeck 2012; da Silva et al. 2020; Morris et al. 2023). Recent machine learning models for time series forecasting tend to emphasise multivariate formulations (Nie et al. 2023; Zeng et al. 2023; Liu et al. 2023; Ansari et al. 2024; Das et al. 2024). While such models provide important insights for integrating traditional time series forecasting into deep learning techniques, e.g. capturing seasonality (Lin et al. 2024), frequency-domain methods (Zhou

et al. 2022a,b), and autoregression (Zeng et al. 2023; Nie et al. 2023), they often overlook the benefits of leveraging external information through exogenous indicators.

In fact, empirical evidence suggests that capturing dependencies from external variables can increase the risk of overfitting for complex models (Nie et al. 2023). Moreover, existing methods that capture inter-variable dependencies (Zhang and Yan 2023; Wang et al. 2024a) have, in certain forecasting tasks, been outperformed by models that do not (Lin et al. 2024; Luo and Wang 2024a). However, this can be attributed to the choice of forecasting benchmarks, which, in many occasions, hold strong seasonal patterns (Lin et al. 2024; see also Appendix E), that can be effectively modelled using simpler autoregressive methods (Zeng et al. 2023; Lin et al. 2024). Additionally, models that do not capture inter-variable dependencies benefit from learning using longer look-back windows without significantly increasing modelling complexity and computational cost (Han, Ye, and Zhan 2023), making them less likely to overfit. Nevertheless, these empirical outcomes are not necessarily valid for multivariable forecasting scenarios, especially for tasks where exogenous variables have strong predictive power. In such cases, explicitly modelling interactions across variables facilitates better performance (Wang et al. 2024b; Shu and Lamos 2025).

Prior work has explored various ways in capturing inter-variable dependencies for multivariate or MTS forecasting tasks. Methods applying attention (Vaswani et al. 2017) across variables (Liu et al. 2024; Ilbert et al. 2024) can capture nonlinear dependencies, but disrupt temporal information by embedding sequences along the time dimension. Crossformer (Zhang and Yan 2023) attempts to rectify this by proposing a modified transformer structure where input is split into time series patches allowing to capture dependencies across both the time and variable dimensions at the sub-sequence level. ModernTCN (Luo and Wang 2024b), on the other hand, uses a convolutional kernel over both the variable and time dimension to jointly capture the inter- and intra-variable dependencies. However, both Crossformer and ModernTCN suffer from GPU computational overhead as the number of exogenous variables or the length of the input time series increases (Shu and Lamos 2025; Zhou et al. 2024). TimeXer (Wang et al. 2024b) models dependencies between exogenous variables and the target variable separately, and then joins their learned embeddings with a cross-attention

module. DeformTime (Shu and Lamos 2025) yields superior results by using deformable attention to incorporate information from exogenous variables at different time steps. However, it only captures inter-variable dependencies within the reception field of a convolutional kernel, which limits the exploration of a wider range of exogenous variables.

Frequency-based analysis techniques, including those using Fourier (Bochner 1953; Sorensen et al. 1987) or wavelet transform (Varanini et al. 1997; Zhang et al. 2003; Arneodo, Grasseau, and Holschneider 1988; Farge et al. 1992; Yu, Guo, and Sano 2024), has been widely used in statistical approaches for identifying periodic patterns (Priestley 2018) as well as in machine learning methods for forecasting (Lange, Brunton, and Kutz 2021; Zhou et al. 2022a,b; Piao et al. 2024). These methods compress temporal information (due to the Fourier transform) while some of them focus on capturing intra-variable dependencies (Zhou et al. 2022a,b). Wavelet transform, albeit dependent on the chosen mother wavelet (De Moortel, Munday, and Hood 2004; Ngui et al. 2013), can maintain both time and frequency information. In the frequency domain, spectral coherence (White and Boashash 1990) serves as a powerful tool for capturing the correlation between variables at different frequencies (Stein, French, and Holden 1972; Mima and Hallett 1999). To improve input projections into the frequency domain, Lange, Brunton, and Kutz (2021) used a Koopman operator (Mezić 2005; Rowley et al. 2009; Avila and Mezić 2020; Liu et al. 2023), i.e. a spectral function that enables linear modelling of nonlinear changes to better address nonlinearities. More recently, AdaWaveNet (Yu, Guo, and Sano 2024) decomposed the input series into seasonal and long-term components, and used the wavelet transform to capture periodic information. This method accounts for inter-variable dependencies only over the seasonal components.

Motivated by the aforementioned remarks, we propose **Spectral Operator Neural Network (Sonnet)**, a model that captures MTS dependencies in the spectral domain using a learnable wavelet transform. Sonnet captures both intra- and inter-variable dependencies using a novel frequency-domain **Multivariable Coherence Attention (MVCA)** layer. Furthermore, it deploys a learnable Koopman operator for linearised transitions of temporal states (Li et al. 2020). MVCA demonstrates stand-alone effectiveness when integrated into existing architectures, outperforming naïve attention and other modified attention mechanisms. Our key contributions are:

1. We propose **Sonnet**, a novel neural network architecture for MTS forecasting that captures inter-variable dependencies via adaptable time-frequency spectral operators while enforcing stability through learnable Koopman dynamics.
2. We introduce **MVCA**, an attention mechanism designed to model interactions between variables by leveraging their spectral coherence, a frequency-domain measurement of dependency. Unlike conventional self-attention, which computes pairwise similarity via dot products, MVCA captures temporal relationships through their cross-spectral density with the inclusion of frequency information from all variables, enhancing variable dependency modelling for MTS tasks.
3. We assess forecasting accuracy on carefully curated

MTS data sets, including established benchmarks complemented by tasks on weather forecasting, influenza prevalence, electricity consumption, and energy prices. The weather prediction and influenza prevalence tasks support a more thorough evaluation as they include a substantial amount of exogenous variables, multiple years (≥ 10) for training, and multiple test periods (≥ 3).

4. Sonnet reduces mean absolute error (MAE) by 2.2% on average compared to the most performant baseline model (stat. sig., $p < 10^{-3}$). In the more challenging tasks of influenza and weather modelling, MAE is reduced by 3.5% and 2%, respectively. Performance gains persist as the forecasting horizon increases, demonstrating the effectiveness of Sonnet in longer-term forecasting.

2 MTS Forecasting Task Definition

We focus exclusively on **multivariable time series (MTS)** forecasting, whereby multiple input variables are used to predict a single target variable. We note that although for some baseline models multiple output variables may be present (**multivariate** forecasting), our evaluation is restricted to the prediction of one specific output. All models are trained and evaluated under a rolling window setup, with a look-back window and a forecasting horizon of L and H time steps, respectively. At each time step t , the C observed exogenous variables over L past time steps, $\{t-L+1, \dots, t\}$, are captured in an input matrix $\mathbf{X}_t \in \mathbb{R}^{L \times C}$. The autoregressive signal for the target (endogenous) variable is denoted by $\mathbf{y}_{t-\delta} \in \mathbb{R}^L$; this encompasses time steps $\{t-\delta-L+1, \dots, t-\delta\}$, where $\delta \in \mathbb{N}_0$ is an optional delay applied when the target variable is observed with a temporal lag. We capture the exogenous and endogenous input variables in $\mathbf{Z}_t = [\mathbf{X}_t, \mathbf{y}_{t-\delta}] \in \mathbb{R}^{L \times (C+1)}$. The goal is to predict the target variable at time step $t+H$, where H denotes the forecasting horizon. Hence, the output of a forecasting model is denoted by $\mathbf{y}_{t+H} \in \mathbb{R}^H$ and holds forecasts for time steps $\{t+1, \dots, t+H-1, t+H\}$. For models that conduct multivariate forecasting, the output includes predictions for all covariates and hence is denoted by $\mathbf{Y}_{t+H} = [\mathbf{X}_{t+H}, \mathbf{y}_{t+H}] \in \mathbb{R}^{H \times (C+1)}$. The forecasting task is to learn $f: \mathbf{Z}_t \rightarrow \mathbf{y}_{t+H}$ or $\mathbf{Y}_{t+H}$. Performance is measured based on the endogenous forecast at time step $t+H$, i.e. the last (temporally) element of $\mathbf{y}_{t+H}$, $y_{t+H} \in \mathbb{R}$. For notational simplicity, we omit temporal subscripts and use $\mathbf{Z}$ for $\mathbf{Z}_t$, $\mathbf{y}$ for $\mathbf{y}_{t-\delta}$, and $\mathbf{X}$ for $\mathbf{X}_t$.

3 Spectral Coherence with Sonnet

In this section, we provide a detailed description of our proposed model, Sonnet. It operates in the spectral domain via a learnable wavelet transform and introduces a **Multivariable Coherence Attention (MVCA)** module to capture both inter- and intra-variable dependencies. We further use a Koopman projection layer that enables stable temporal evolution via a learned linear operator.

3.1 Joint Embedding of Input Variables

Given an input matrix $\mathbf{Z} \in \mathbb{R}^{L \times (C+1)}$ that consists of the exogenous variables $\mathbf{X} \in \mathbb{R}^{L \times C}$ and endogenous variable $\mathbf{y} \in \mathbb{R}^L$, we first obtain the embeddings of $\mathbf{X}$ and $\mathbf{y}$ independently.

These are denoted by $\mathbf{E}_x \in \mathbb{R}^{L \times \alpha d}$ and $\mathbf{E}_y \in \mathbb{R}^{L \times (1-\alpha)d}$. They are derived using learnable weight matrices $\mathbf{W}_x \in \mathbb{R}^{C \times \alpha d}$ and $\mathbf{W}_y \in \mathbb{R}^{1 \times (1-\alpha)d}$, as in $\mathbf{E}_x = \mathbf{X}\mathbf{W}_x$, where d is the embedding dimension and $\alpha \in [0, 1]$ a hyperparameter that controls the projected dimensionality of $\mathbf{X}$ and $\mathbf{y}$ in the final embedding.¹ By concatenating along the feature dimension, we obtain the final embedding $\mathbf{E} = [\mathbf{E}_x, \mathbf{E}_y] \in \mathbb{R}^{L \times d}$.

3.2 Learnable Wavelet Transform

We then transform the time series embedding into the wavelet space that contains both time and frequency information, to capture both fine-grained details and overall trends across various temporal resolutions (Mallat 1989; Daubechies 1990). Specifically, after obtaining the input time series embedding $\mathbf{E}$, we first define a set of K learnable wavelet transformations (also referred to as atoms), with the k -th atom held in a matrix $\mathbf{M}_k \in \mathbb{R}^{d \times L}$ derived by

$$\mathbf{M}_k = \exp(-\mathbf{w}_\alpha \mathbf{t}^2) \times \cos(\mathbf{w}_\beta \mathbf{t} + \mathbf{w}_\gamma \mathbf{t}^2), \quad (1)$$

where $\mathbf{t} \in \mathbb{R}^L$ is a row vector capturing normalised time steps with $\mathbf{t}_i = i/(L-1)$ for $i = 0, \dots, L-1$, and $\mathbf{w}_\alpha, \mathbf{w}_\beta$, and $\mathbf{w}_\gamma \in \mathbb{R}^d$ are learnable weight vectors that control the shape of the wavelet, each initialised randomly from a normal distribution. In particular, $\mathbf{w}_\alpha$ controls the width of the Gaussian envelope, and $\mathbf{w}_\beta, \mathbf{w}_\gamma$ respectively determine the linear or quadratic frequency modulation of the generated cosine waveforms. This formulation enables the atoms to adapt to localised time-frequency structures in the data. The time series embedding $\mathbf{E} \in \mathbb{R}^{L \times d}$ is then transformed into the wavelet space by projecting it onto each of the wavelet atoms $\mathbf{M}_k \in \mathbb{R}^{d \times L}$ using $\mathbf{P}_k = \mathbf{E} \odot \mathbf{M}_k^\top$, where $\mathbf{P}_k \in \mathbb{R}^{L \times d}$ denotes the embedding's projection for the k -th atom, and $\odot$ is element-wise multiplication. The transformed wavelet across all K atoms is denoted by $\mathbf{P} \in \mathbb{R}^{K \times L \times d}$. The aforementioned steps help to preserve temporal structure while decomposing the input into multi-resolution time-frequency components using adaptive wavelet transforms that can capture both short- and long-term patterns in the data.

3.3 Multivariable Coherence Attention (MVCA)

In MTS forecasting, input variables can be both auto-correlated (to their own past values) and cross-correlated (to each other). To improve learning from these correlations, we propose MVCA, a module that supplements the standard attention. MVCA can capture inter- and intra-variable dependencies within the frequency domain using spectral density coherence. Its premise is that variables with a higher spectral coherence should contribute more to the attention output. MVCA can be used in place of any naïve attention module used in a forecasting method.

Given the obtained input embedding matrix in wavelet space $\mathbf{P} \in \mathbb{R}^{K \times L \times d}$, where L and d are the time and variable dimension, K is the number of wavelets after transformation,

¹We make sure that both products αd and $(1-\alpha)d \in \mathbb{N}$ to avoid dimension mismatch caused by rounding. For $\alpha = 0$, the forecasting is based on the historical values of the target variable only, collapsing to an autoregressive setting. For $\alpha = 1$, forecasting depends entirely on the exogenous variables.

we first linearly project it to query, key and value embeddings, denoted as $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V} \in \mathbb{R}^{K \times L \times d}$, using weight matrices $\mathbf{W}_q, \mathbf{W}_k$, and $\mathbf{W}_v \in \mathbb{R}^{d \times d}$ respectively, as in $\mathbf{Q} = \mathbf{P}\mathbf{W}_q$. We then consider the embeddings from different wavelet transformations (indexed by k) as separate attention heads. Specifically, for each attention head, we obtain the sub-tensor of $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$, denoted as $\mathbf{Q}_h, \mathbf{K}_h$, and $\mathbf{V}_h \in \mathbb{R}^{L \times d}$, as the query, key, and value embeddings. Therefore, each head captures a distinct subspace of the original embedding, where subspaces correspond to different wavelet transformations of the input. Including a multi-head structure enables the model to learn diverse dependencies in parallel (Vaswani et al. 2017).

We then apply a Fast Fourier Transform (FFT) along the variable dimension (Dudgeon and Mersereau 1984) of each transformer head to transfer the query and key embeddings to the frequency domain, i.e. $\mathbf{Q}_f = \text{FFT}(\mathbf{Q}_h)$ and $\mathbf{K}_f = \text{FFT}(\mathbf{K}_h)$, with both $\mathbf{Q}_f$ and $\mathbf{K}_f \in \mathbb{C}^{L \times \ell}$, where $\ell = \lfloor \frac{d}{2} \rfloor + 1$. Each frequency bin in the transformed data ($\mathbf{Q}_f, \mathbf{K}_f$) contains information from all original inputs to the FFT (Lee-Thorp et al. 2022), capturing both fine-grained (high-frequency) and global (low-frequency) patterns. The cross-spectral density $\mathbf{P}_{qk} \in \mathbb{C}^{L \times \ell}$ and power-spectral densities $\mathbf{P}_{qq}, \mathbf{P}_{kk} \in \mathbb{R}^{L \times \ell}$ are obtained as follows:

$$\mathbf{P}_{qk} = \mathbf{Q}_f \odot \mathbf{K}_f^*, \mathbf{P}_{qq} = \mathbf{Q}_f \odot \mathbf{Q}_f^*, \mathbf{P}_{kk} = \mathbf{K}_f \odot \mathbf{K}_f^*, \quad (2)$$

where * denotes the complex conjugate. We average along the second dimension to obtain $\bar{\mathbf{P}}_{qk} \in \mathbb{C}^L$, $\bar{\mathbf{P}}_{qq}$, and $\bar{\mathbf{P}}_{kk} \in \mathbb{R}^L$ (see also Appendix B). The normalised spectral coherence $\mathbf{C}_{qk} \in \mathbb{R}^L$ is then computed using

$$\mathbf{C}_{qk} = |\bar{\mathbf{P}}_{qk}|^2 / (\bar{\mathbf{P}}_{qq} \cdot \bar{\mathbf{P}}_{kk} + \epsilon), \quad (3)$$

where $\epsilon = 10^{-6}$ mitigates division by 0. $\mathbf{C}_{qk}$ captures the linear dependency between sequences across frequency bands. Therefore, the coherence here assigns higher importance to the time steps where the query and key hold more similar averaged values across multiple frequencies.

Following the common design of attention layers (Vaswani et al. 2017), we scale, normalise, and regularise $\mathbf{C}_{qk}$, i.e. $\mathbf{A}_h = \text{Dropout}(\text{Softmax}(\mathbf{C}_{qk}/\sqrt{d}))$. The attention weights $\mathbf{A}_h \in \mathbb{R}^L$ of each head are first broadcast along the feature dimension to form $\mathbf{A} \in \mathbb{R}^{L \times d}$, which is then multiplied with the value representations $\mathbf{V}_h$ (element-wise) to produce head-specific outputs, $\mathbf{O}_h \in \mathbb{R}^{L \times d} = \mathbf{A} \odot \mathbf{V}_h$. We concatenate these outputs across all heads to obtain $\mathbf{O}_r \in \mathbb{R}^{K \times L \times d}$. We then use a 2-layer perceptron (MLP, d -dimensional layers) with Gaussian Error Linear Unit (GELU) activation to further capture nonlinearities. This is connected to a residual layer to form the output $\mathbf{O}_m \in \mathbb{R}^{K \times L \times d} = \mathbf{O}_r + \text{MLP}(\mathbf{O}_r)$. We then multiply $\mathbf{O}_m$ with a weight matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times d}$ to obtain the output of MVCA, $\mathbf{O} \in \mathbb{R}^{K \times L \times d} = \mathbf{O}_m \mathbf{W}_{\text{out}}$.

3.4 Koopman-Guided Spectrum Evolvement

Motivated by Koopman operator theory (Mezić 2005; Rowley et al. 2009; Avila and Mezić 2020), which offers a framework for modelling nonlinear dynamics, we introduce a layer to capture the temporal evolution of time-frequency patterns in

wavelet space. We aim to learn a Koopman operator with K dimensions in the transformed space. To obtain the operator $\mathbf{K} \in \mathbb{C}^{K \times K}$, we first initialise a learnable complex-valued matrix $\mathbf{S} \in \mathbb{C}^{K \times K}$. At each forward pass, we apply QR decomposition to $\mathbf{S}$, and retain only the resulting unitary matrix $\mathbf{U} \in \mathbb{C}^{K \times K}$, i.e. $\mathbf{U} \leftarrow \text{QR}(\mathbf{S})$ s.t. $\mathbf{U}^\dagger \mathbf{U} = \mathbf{I}$, where $\mathbf{U}^\dagger$ is the conjugate transpose of $\mathbf{U}$. Multiplying with $\mathbf{U}$ therefore prevents data amplification or distortion.

We then initialise a learnable vector $\mathbf{p} \in \mathbb{R}^K$, where the k -th element, p_k , controls the temporal evolution (phase transformation angle) of the k -th transformation. All elements in the vector are then mapped into complex numbers, obtaining $\mathbf{v} \in \mathbb{C}^K$, where $v_k = e^{ip_k}$. We use $\mathbf{v}$ to form a diagonal matrix $\mathbf{D} \in \mathbb{C}^{K \times K} = \text{diag}(\mathbf{v})$. The Koopman operator, $\mathbf{K}$, is then given by $\mathbf{K} = \mathbf{U} \mathbf{D} \mathbf{U}^\dagger$. We use $\mathbf{K}$ to model the temporal evolution of the MVCA embedding, $\mathbf{O}$, after converting it to a complex form, $\mathbf{O}_c \in \mathbb{C}^{K \times L \times d}$ (the imaginary part is set to $i0$ to perform complex-valued transformations without altering the original embedding), given by $\mathbf{O}_l = \mathbf{K} \times \mathbf{O}_c$, where $\mathbf{O}_l \in \mathbb{C}^{K \times L \times d}$ is the evolved embedding after multiplication with $\mathbf{K}$. A conventional Koopman framework models temporal evolution recursively at each time step (Lusch, Kutz, and Brunton 2018; Avila and Mezić 2020). We instead choose to apply Koopman transformation in the frequency domain with one forward pass, which is equivalent to learning a direct transformation from the input to the output. This global projection reduces the accumulation of sequential errors while maintaining robustness in training.

3.5 Sequence Reconstruction from Wavelets

The inverse transformation reconstructs the original sequence by aggregating weights from each wavelet atom. Given the evolved state $\mathbf{O}_l$, we first obtain its real part denoted as $\mathbf{O}_r \in \mathbb{R}^{K \times L \times d}$. For each wavelet atom indexed by k , let $\mathbf{O}_k \in \mathbb{R}^{L \times d}$ denote the corresponding slice of $\mathbf{O}_r$. We then multiply it with the wavelet atom $\mathbf{M}_k \in \mathbb{R}^{d \times L}$, i.e. $\mathbf{R}_k = \mathbf{O}_k \odot \mathbf{M}_k^\top$. $\mathbf{R}_k \in \mathbb{R}^{L \times d}$ denotes the series reconstruction from the k -th atom. This operation is conducted over all K atoms. Joining all the reconstructed series forms a $K \times L \times d$ matrix. We sum over its first dimension to obtain the reconstructed embedding, denoted as $\mathbf{R} \in \mathbb{R}^{L \times d}$.

3.6 Convolutional Decoder

Finally, a 3-layer convolutional decoder transforms the learned representation. It comprises 3 1-dimensional convolutional layers with GELU activations between every 2 layers. Each layer uses kernel sizes $[5, 3, 3]$ and paddings $[2, 1, 1]$ respectively, followed by an adaptive average pooling layer at the end. The dimension of each layer is $[H \times 4, H \times 2, H]$, producing the final sequence representation $\mathbf{Z}_{\text{out}} \in \mathbb{R}^{H \times H}$, in accordance with the time steps of the target forecasting horizon. The result is then linearly projected using a weight vector $\mathbf{W}_z \in \mathbb{R}^H$ to generate the final output, as in $\hat{\mathbf{y}} \in \mathbb{R}^H = \mathbf{Z}_{\text{out}} \mathbf{W}_z$, in accordance with the dimensionality of the target variable.

4 Results

We assess forecasting accuracy using an expanded collection of data sets and tasks, to overcome potential biases present in

the current literature. We first compare Sonnet against other competitive baseline models. We then investigate the role of attention mechanisms in time series forecasting models by removing or replacing naïve transformers with more advanced variants, including the proposed MVCA module. We also provide an ablation study of the key components of Sonnet and seed control in Appendix E.

4.1 Experiment Settings

We conduct experiments over 12 real-world data sets. This includes 2 established benchmarks from prior papers (Zhou et al. 2021; Zeng et al. 2023), specifically the ETTh1 and ETTh2 data sets, which contain hourly electricity transformer temperature forecasting. Oil temperature is our target variable, with the remaining indicators considered as exogenous variables, following Wang et al. (2024b). We also use 2 data sets from the Darts library (Herzen et al. 2022), predicting hourly energy prices (ENER) and electricity (low-voltage) consumption (ELEC). In addition, we form weather data sets extracted from the WeatherBench repository (Rasp et al. 2020) for 5 cities from diverse geographical locations, namely London (WEA-LD), New York (WEA-NY), Hong Kong (WEA-HK), Cape Town (WEA-CT), and Singapore (WEA-SG), to support a more inclusive analysis. For each city, we sample data from its nearest grid point and obtain 5 climate indicators. We provide spatial context by including data from its eight surrounding grid points (3×3 grid) as additional climate exogenous variables. Following prior work on global climate forecasting (Verma, Heinonen, and Garg 2024), we resample the data to a temporal resolution of 6 hours. The forecasting target is the 850 hPa (T850) temperature, a key indicator for climate modelling (Scherrer et al. 2004; Hamill and Whitaker 2007). Finally, we include influenza-like illness (ILI) rate forecasting tasks (as in (Shu and Lamos 2025)) in 3 locations, England (ILI-ENG), and in U.S. Health & Human Services (HHS) Regions 2 (ILI-US2) and 9 (ILI-US9). In the ILI tasks, frequency time series of web searches are included as exogenous predictors. More information about the data sets is provided in Appendix A.

For the ETT tasks, we set the forecasting horizon (H) to $\{96, 192, 336, 720\}$ time steps, and use a single test set of consecutive unseen instances adopting the evaluation settings in prior work (Nie et al. 2023; Liu et al. 2024). For ENER we set $H = \{24, 48, 72, 168\}$ hours ahead and use 1 year (2018) for testing. For ELEC, we use the last 2 years (2020, '21) as 2 distinct test seasons, and set H to $\{12, 24, 36\}$ hours. For the WEA tasks, we form 3 test sets (years 2016, '17, '18), and set H to $\{4, 12, 28, 120\}$ time steps corresponding to $\{1, 3, 7, 30\}$ days. Following the same setup as in (Shu and Lamos 2025), for the ILI forecasting task, we test models on 4 consecutive influenza seasons (2015/16 to 2018/19), each time training a new model on data from previous seasons. We set $H = \{7, 14, 21, 28\}$ days (more details in Appendix D).

We compare Sonnet to 8 competitive forecasting models that, to the best of our knowledge, form the current SOTA methods: DLinear (Zeng et al. 2023), Crossformer (Zhang and Yan 2023), iTransformer (Liu et al. 2024), PatchTST (Nie et al. 2023), TimeXer (Wang et al. 2024b), Samformer (Ilbert et al. 2024), ModernTCN (Luo and Wang 2024b), and

Task	Sonnet		DeformTime		ModernTCN		Samformer		TimeXer		PatchTST		iTransformer		Crossformer		DLinear		
	H	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%
ELEC	12	0.1040	24.95	0.1162	26.68	0.1596	36.81	0.2336	53.84	0.1287	28.45	0.1419	30.76	0.1468	31.91	0.1513	30.35	0.3307	71.51
	24	0.1203	27.70	0.1359	<u>30.11</u>	0.1806	39.70	0.2520	56.39	0.1586	33.20	0.1545	34.04	0.1588	35.40	0.1873	35.60	0.3369	73.23
	36	0.1389	30.21	0.1729	35.81	0.2065	43.79	0.2866	62.40	0.1785	35.33	<u>0.1659</u>	<u>33.85</u>	0.1791	38.18	0.2273	39.36	0.3908	82.38
ENER	24	0.3621	65.67	0.3717	66.92	0.3752	66.53	<u>0.3703</u>	66.01	0.3733	69.59	0.3717	<u>65.89</u>	0.3838	68.36	0.3716	67.44	0.4307	76.49
	48	0.4120	71.07	0.4299	74.59	0.4154	71.74	0.4294	72.34	0.4404	77.61	0.4467	75.60	0.4386	75.63	0.4647	81.79	0.5138	89.79
	72	0.4036	70.52	0.4217	74.50	<u>0.4089</u>	<u>70.84</u>	0.4353	73.40	0.4246	74.70	0.4303	73.90	0.4276	74.19	0.4473	78.62	0.5124	89.26
168	0.3980	68.63	0.4399	78.98	0.4497	74.86	0.4243	<u>72.28</u>	<u>0.4493</u>	76.76	0.4872	81.34	0.4352	74.11	<u>0.4168</u>	76.76	0.5070	87.87	
ETTh1	96	0.2145	16.01	0.1941	14.96	0.2047	15.66	0.2047	15.73	0.2135	16.03	<u>0.2017</u>	<u>15.41</u>	0.2052	15.46	0.2126	16.52	0.2599	20.82
	192	0.2330	17.61	0.2116	16.08	0.2417	18.32	<u>0.2307</u>	17.64	0.2322	<u>16.96</u>	<u>0.2409</u>	18.29	0.2429	18.13	0.2820	21.63	0.3798	31.78
	336	<u>0.2392</u>	18.91	0.2158	16.27	0.2415	18.52	0.2523	18.94	0.2414	<u>17.34</u>	0.2559	19.29	0.2593	19.11	0.2947	22.65	0.6328	58.34
720	<u>0.2768</u>	<u>19.73</u>	0.2862	21.81	0.2785	20.44	0.3026	23.21	0.2617	18.58	0.3087	23.89	0.2886	22.05	0.3350	24.84	0.7563	69.52	
ETTh2	96	0.3098	33.04	0.3121	40.07	0.3199	40.68	0.3312	40.16	0.3346	41.04	0.3145	39.25	0.3420	42.41	0.3486	40.71	0.3349	41.68
	192	<u>0.3742</u>	<u>39.58</u>	0.3281	37.90	0.3887	47.08	0.3874	45.37	0.4154	47.07	0.3839	45.45	0.4233	47.44	0.4035	43.16	0.4084	50.67
	336	<u>0.3689</u>	<u>39.48</u>	0.3450	37.00	0.3904	50.54	0.4083	46.41	0.4041	42.26	0.4018	46.77	0.4332	45.95	0.4487	49.44	0.4710	55.53
720	<u>0.4335</u>	<u>51.62</u>	0.3640	34.99	0.5728	63.04	0.5198	58.44	0.5135	56.17	0.4960	55.27	0.4565	45.40	0.5832	61.45	0.7981	94.67	
ILI-ENG	7	1.4791	21.84	1.6417	28.60	1.9489	28.27	2.3475	28.31	2.8084	33.66	2.3115	27.61	2.3084	26.38	1.8698	<u>25.70</u>	2.8214	43.02
	14	1.9225	25.77	<u>2.2308</u>	33.98	2.7050	36.01	3.0290	36.63	3.4937	41.88	3.2547	37.76	3.2301	36.67	2.6543	<u>30.97</u>	3.7922	55.29
	21	2.5101	36.53	<u>2.6500</u>	32.70	3.0400	40.02	4.4980	54.41	4.3337	51.57	4.3192	51.11	4.2347	48.93	3.0014	40.57	4.4739	61.25
28	<u>2.7481</u>	<u>36.95</u>	2.7228	40.44	3.3611	47.87	5.1598	60.78	4.9013	61.60	4.9964	59.60	4.8125	55.35	3.1983	46.15	5.0347	67.75	
ILI-US2	7	0.3806	14.89	0.4122	<u>16.01</u>	0.4398	16.55	0.6495	24.21	0.6083	23.38	0.7097	24.51	0.6507	23.24	0.4400	16.46	0.7355	27.94
	14	0.4491	<u>18.38</u>	<u>0.4752</u>	17.73	0.5279	20.22	0.7696	30.16	0.7725	29.07	0.8635	30.11	0.7896	28.17	0.5852	20.98	0.8435	32.22
	21	0.5326	20.64	<u>0.5425</u>	<u>22.12</u>	0.5781	23.85	0.8374	31.42	0.8243	31.46	1.0286	36.70	0.8042	30.03	0.6245	22.29	0.9124	34.93
28	0.5788	21.15	0.5538	<u>22.25</u>	<u>0.5710</u>	23.66	0.9389	36.80	0.9074	34.72	1.1525	42.61	0.9619	36.75	0.6512	23.47	0.9999	38.46	
ILI-US9	7	0.2668	<u>12.98</u>	0.2622	12.26	0.2899	14.17	0.4025	19.39	0.3813	18.21	0.4116	19.34	0.4057	18.57	0.3149	14.44	0.4675	23.47
	14	0.2806	13.10	<u>0.3084</u>	<u>13.80</u>	0.3417	15.29	0.5257	24.50	0.4665	22.14	0.5020	24.09	0.4702	22.44	0.3571	17.23	0.5467	27.35
	21	0.3179	14.11	0.3179	<u>14.23</u>	0.3710	15.43	0.5415	24.22	0.5715	27.43	0.5935	29.40	0.5106	24.11	0.3418	15.90	0.6001	29.66
28	<u>0.3675</u>	16.53	0.3532	15.75	0.3940	17.19	0.6050	27.95	0.6555	31.32	0.6665	33.35	0.6498	31.05	0.3747	<u>16.44</u>	0.6564	32.16	
WEA-CT	4	1.6240	<u>9.63</u>	1.7600	10.41	1.8752	11.05	2.2265	13.09	2.2376	13.14	3.5004	20.17	2.1906	12.83	<u>1.6382</u>	9.62	3.1483	18.22
	12	3.5432	20.38	<u>3.5681</u>	<u>20.46</u>	3.7761	21.67	4.0444	23.14	3.7241	21.31	4.1910	23.99	3.9741	22.83	3.5932	20.62	4.0265	22.94
	28	3.7277	21.32	<u>3.7601</u>	<u>21.49</u>	3.9399	22.36	3.9239	22.37	3.8325	21.87	3.9405	22.48	3.9584	22.60	3.8061	21.72	3.9254	22.37
120	3.7373	21.35	3.8040	21.67	4.0889	23.54	3.9018	22.30	3.8412	21.88	4.0547	23.02	3.9473	22.49	<u>3.7659</u>	<u>21.49</u>	4.0570	23.05	
WEA-HK	4	0.6389	4.05	0.6804	4.32	0.7004	4.42	0.8097	5.10	0.8648	5.52	1.1488	7.15	0.8048	5.08	<u>0.6488</u>	<u>4.11</u>	0.9898	6.21
	12	1.2355	7.74	<u>1.2786</u>	<u>7.99</u>	1.3555	8.43	1.4006	8.70	1.3027	8.13	1.5825	9.77	1.4225	8.86	1.2896	8.08	1.5464	9.62
	28	1.4135	8.83	<u>1.4746</u>	<u>9.16</u>	1.5866	9.82	1.6226	10.05	1.5115	9.39	1.6441	10.16	1.6356	10.08	1.5282	9.45	1.6232	10.06
120	<u>1.5469</u>	<u>9.58</u>	1.5399	9.45	1.6326	10.22	1.8843	12.06	1.7092	10.55	2.0084	12.84	1.6745	10.45	1.5931	9.73	1.8329	11.49	
WEA-LD	4	1.7231	15.16	1.8753	16.31	1.9456	16.88	2.1537	18.53	2.2628	19.25	2.7602	22.93	2.1509	18.55	<u>1.7447</u>	<u>15.26</u>	2.5065	20.94
	12	2.9589	23.88	<u>3.0214</u>	<u>24.15</u>	3.2056	25.58	3.3070	26.61	3.1625	25.25	3.5406	28.33	3.3622	27.36	3.0492	24.35	3.3927	26.70
	28	3.2161	25.49	<u>3.2724</u>	<u>25.84</u>	3.5067	27.48	3.5841	27.97	3.4672	27.23	3.7365	29.42	3.6884	29.14	3.3048	25.88	3.6073	28.16
120	3.2464	25.82	<u>3.2973</u>	<u>26.17</u>	3.8434	29.92	3.8420	30.32	3.6557	28.52	4.2344	32.40	3.8518	30.36	3.3935	26.75	3.9640	30.35	
WEA-NY	4	1.2716	11.94	1.4028	13.03	1.4154	12.95	1.6003	14.44	1.7290	15.57	2.1644	19.24	1.6066	14.40	<u>1.2935</u>	<u>12.24</u>	1.9782	17.88
	12	<u>2.4476</u>	21.23	2.4453	21.04	2.6221	22.93	2.7069	23.57	2.6537	22.85	2.8592	24.81	2.6609	23.09	2.4494	<u>21.11</u>	2.9507	25.29
	28	2.6744	23.10	<u>2.7450</u>	<u>23.20</u>	2.9336	25.37	3.0347	26.18	2.8775	24.48	3.0956	27.01	3.0204	25.30	2.7830	23.66	3.2099	27.07
120	2.7135	23.15	<u>2.8224</u>	<u>23.86</u>	3.3000	28.05	3.5289	33.01	3.1501	26.55	3.4086	28.78	3.1029	27.74	2.9615	24.69	3.6129	29.75	
WEA-SG	4	0.3444	1.25	0.3557	1.30	0.3624	1.32	0.3925	1.43	0.3801	1.38	0.4238	1.54	0.3868	1.41	<u>0.3532</u>	<u>1.29</u>	0.4048	1.47
	12	0.4160	1.51	0.4256	<u>1.55</u>	0.4493	1.64	0.4784	1.74	0.4447	1.62	0.4992	1.82	0.4662	1.70	0.4359	1.59	0.4764	1.73
	28	0.4653	1.69	<u>0.4875</u>	<u>1.77</u>	0.5196	1.89	0.5421	1.97	0.4974	1.81	0.5542	2.02	0.5307	1.93	0.5003	1.82	0.5284	1.92
120	0.4830	1.76	<u>0.5050</u>	<u>1.83</u>	0.5321	1.94	0.5221	1.90	0.5215	1.90	0.5357	1.95	0.5233	1.90	0.5212	1.89	0.5328	1.94	

Table 1: Forecasting accuracy results across all tasks, methods, and forecasting horizons (H). For the ELEC/ILI/WEA tasks, we report the average performance across 2, 4, and 3 test sets, respectively (detailed breakdowns in Appendix E.1). $\epsilon\%$ denotes sMAPE. Best results are **bolded**, second best underlined. Grey background denotes the model does not outperform persistence.

DeformTime (Shu and Lamos 2025), with TimeXer and DeformTime being designed for multivariable forecasting. We also include a naïve baseline which can either be a seasonal or a standard persistence model, depending on the presence of strong seasonality. For all tasks (except ETT), we conduct hyperparameter tuning for all models. For the ETT tasks, we adopt settings from the official repositories of methods

(except for Crossformer, which did not provide this). Appendix C has further details about the baseline models.

4.2 Forecasting Accuracy of Sonnet

Prediction accuracy for all tasks is enumerated in Table 1, using Mean Absolute Error (MAE) and symmetric Mean Absolute Percentage Error (sMAPE or $\epsilon\%$) as the main evaluation

	Attention	$H = 7$ days			$H = 14$ days			$H = 21$ days			$H = 28$ days		
		ENG	US2	US9	ENG	US2	US9	ENG	US2	US9	ENG	US2	US9
iTransformer	—	2.3084	0.6507	0.4057	3.2301	0.7896	0.4702	4.2347	0.8042	0.5106	4.8125	0.9619	0.6498
	$\neg$ Attn	2.3011	0.7242	0.4475	3.2557	0.8696	0.5344	4.4729	1.0105	0.6207	5.1864	1.1667	0.7179
	FNet	2.5534	0.7665	0.4741	3.5337	0.8746	0.5550	4.6970	1.0006	0.6066	5.3153	1.1118	0.7139
	FED	3.6150	0.9286	0.5613	4.3098	1.0750	0.6527	5.8596	1.4140	0.8669	6.4196	1.5167	0.9397
	VDAB	2.4287	0.6772	0.4126	3.3093	0.8274	0.4982	3.9406	0.8485	<u>0.4992</u>	4.4975	0.9068	<u>0.5790</u>
	MVCA	2.2707	0.5483	0.3581	3.1233	0.7267	0.4454	<u>4.1780</u>	0.7578	0.4880	4.8705	0.8885	0.5409
SamFormer	—	2.3475	0.6495	0.4025	3.0290	0.7696	0.5257	4.4980	0.8374	0.5415	5.1598	0.9389	0.6050
	$\neg$ Attn	2.5010	0.8245	0.5131	3.3015	0.9679	0.6047	4.1830	1.1396	0.7389	5.0136	1.2391	0.8347
	FNet	2.4812	0.7765	0.4699	3.2054	0.9202	0.5865	4.3909	0.9266	0.6443	5.0592	1.0983	0.7149
	FED	3.5217	0.9140	0.5599	4.2109	1.0519	0.6508	5.7800	1.3439	0.8519	5.9809	1.4348	0.9190
	VDAB	2.0406	0.5755	0.3651	2.9525	0.7281	0.4505	4.1080	0.7883	0.4861	4.9015	0.8617	0.5605
	MVCA	2.1365	0.5514	0.3609	3.2046	0.6935	0.4379	3.8033	0.7294	<u>0.4975</u>	4.7438	0.8159	0.5345
PatchTST	—	2.3115	0.7097	0.4116	3.2547	0.8635	0.5020	4.3192	1.0286	0.5935	4.9964	1.1525	0.6665
	$\neg$ Attn	2.4723	0.7702	0.4585	3.6415	0.9017	0.5551	4.2998	1.0657	0.6472	4.8538	1.1704	0.7313
	FNet	2.8097	0.8033	0.4591	4.0342	0.9322	0.5327	4.7975	1.0341	0.5430	4.9718	1.1087	0.6279
	FED	3.6158	0.9292	0.5614	4.6572	1.0765	0.6542	5.8725	1.4043	0.8671	6.3728	1.4953	0.9306
	VDAB	2.0799	0.5925	0.3820	3.0211	0.7722	0.4413	3.8164	0.8009	0.5159	4.6044	0.8518	0.5801
	MVCA	2.0054	0.5824	0.3705	3.0411	0.7406	0.4291	<u>3.8627</u>	0.7871	0.4746	4.5712	0.8765	0.5491

Table 2: Performance (average MAE across 4 test seasons) of iTransformer, Samformer, and PatchTST on the ILI forecasting tasks (ILI-ENG/US2/US9) with different modifications to the naïve attention mechanism. ‘ $\neg$ Attn’ denotes the removal of the residual attention module, and FNet / FED / VDAB refer to using the attention modules proposed in FNet (2025), FEDformer (2022a), and DeformTime (2025), respectively. Best results are **bolded**, second best underlined.

metrics. For WEA tasks, sMAPE is modified to avoid small values due to Kelvin temperature units (see Appendix D). For ETT and ENER tasks, results are based on a single forecasting season (i.e., one train / test split per forecasting horizon), while we average over multiple seasons for other tasks.

Sonnet exhibits the overall best performance across the explored MTS forecasting tasks. Comparing Sonnet’s performance to the best-performing baseline model for each task and forecasting horizon yields an MAE reduction of 1.1% on average. Ranking-wise, Sonnet is the best-performing model on 34, and the second-best on 9 out of 47 forecasting tasks. Sonnet shows relatively inferior performance on the ETT tasks. However, the ETT data sets have only 6 exogenous variables and cover a limited training time span (2 years). Consequently, they offer a limited spectrum to gain from exploring variable dependencies, as well as insufficient historical context, both of which are elements that Sonnet leverages from. Aside from the ETT tasks, Sonnet reduces MAE by 3.3% on average. Therefore, Sonnet is more effective with more covariates and longer time spans for training. Compared to a specific forecasting model, Sonnet consistently improves accuracy, reducing MAE from 2.2% for the best-performing baseline (DeformTime) to 31.1% for the worst-performing one (DLinear). We note that the MAE performance gain over DeformTime is statistically significant based on a paired t-test across all the 47 forecasting tasks ($p = 5e - 4$). This further highlights the model’s forecasting capacity amongst a wide range of applications.

Focusing on the more challenging forecasting tasks (ILI and WEA), Sonnet outperforms baselines 25 or 26 (out of 32 tasks in total) based on sMAPE or MAE, respectively. The MAE reduction is 3.5% on average for ILI tasks and 2% for WEA tasks. Within these tasks, we also observe consistent comparative performance patterns in the baseline models. Models that do not capture inter-variable dependencies

(DLinear and PatchTST) generally offer lower predictability (Sonnet reduces their MAE by 28.8% and 29.8%), and in most ILI tasks, they cannot surpass the performance of a persistence model. In contrast, baseline models that capture inter-variable dependencies tend to perform better. Among these, those that embed the covariates along the temporal dimension, namely Samformer, TimeXer, and iTransformer, exhibit inferior performance (Sonnet reduces their MAE by 23.9%, 22.1%, and 22.9%), whereas models that preserve temporal order, namely DeformTime, ModernTCN, and Crossformer, achieve stronger results (Sonnet reduces their MAE by 3.4%, 11.1%, and 8%). Hence, in tasks with more informative covariates, preserving temporal structure while modelling inter-variable dependencies is a desirable property. We provide ablation study results in Appendix E.3 that quantify the contribution of different modules to the overall forecasting accuracy of Sonnet. Additional performance evaluation results over the entire output sequence are provided in Appendix E.8.

4.3 Effectiveness of Different Attention Modules in MTS Forecasting

We evaluate the effectiveness of the proposed attention-driven module, MVCA, by integrating it into existing forecasting models (we refer to them as base models) that originally deploy naïve transformer attention. Experiments are conducted on the ILI tasks as these are the most representative of the MTS class we are exploring: they not only contain many exogenous predictors, but also frame hard practical epidemiological modelling problems when $H = 21$ or 28 days ahead. We use Samformer (Ilbert et al. 2024), iTransformer (Liu et al. 2024), and PatchTST (Nie et al. 2023) as base models. For each one of them, we evaluate 5 attention configurations: removing the attention module altogether ($\neg$ Attn), attention with Fourier transformer proposed by FNet (Lee-Thorp

	<i>H</i>	Sonnet		VDAB		FED		FNet		Naïve	
		MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%	MAE	ε%
ILI-ENG	7	1.479	21.8	1.433	20.3	5.991	77.7	1.591	27.7	1.717	25.3
	14	1.923	25.8	1.973	29.4	6.031	78.1	2.084	32.8	2.068	31.4
	21	2.510	36.5	2.774	35.2	6.045	78.2	<u>2.765</u>	39.3	2.781	37.2
	28	2.748	37.0	<u>3.060</u>	<u>44.3</u>	6.051	78.4	3.240	44.5	3.078	50.0
ILI-US2	7	0.381	14.9	0.429	16.6	1.168	45.0	0.415	18.5	0.436	17.5
	14	0.449	18.4	<u>0.500</u>	<u>19.1</u>	1.179	45.5	0.558	23.2	0.532	20.4
	21	0.533	20.6	<u>0.575</u>	20.5	1.195	46.2	0.659	25.3	0.613	23.5
	28	0.579	21.1	0.643	24.3	1.193	46.1	0.705	29.1	0.651	26.1
ILI-US9	7	0.267	13.0	<u>0.277</u>	<u>13.4</u>	0.856	41.0	0.286	14.4	0.301	13.9
	14	0.281	13.1	<u>0.324</u>	15.8	0.856	41.0	0.351	16.1	0.340	<u>14.9</u>
	21	0.318	14.1	<u>0.319</u>	<u>14.5</u>	0.857	41.0	0.379	17.8	0.404	19.2
	28	0.367	16.5	<u>0.375</u>	<u>16.8</u>	0.857	41.0	0.379	18.2	0.411	19.1
WEA-CT	4	1.624	9.6	1.714	10.1	4.359	24.6	<u>1.629</u>	<u>9.6</u>	1.726	10.2
	12	3.543	20.4	3.563	20.5	4.359	24.7	3.588	20.6	<u>3.558</u>	<u>20.5</u>
	28	3.728	21.3	3.760	21.5	4.359	24.6	3.916	22.3	<u>3.758</u>	<u>21.5</u>
	120	<u>3.737</u>	<u>21.4</u>	3.734	21.3	4.356	24.6	3.815	21.7	3.763	21.5
WEA-HK	4	0.639	4.1	0.645	4.1	2.801	17.7	<u>0.639</u>	<u>4.1</u>	0.676	4.3
	12	1.235	7.7	<u>1.279</u>	<u>8.0</u>	2.801	17.7	1.325	8.3	1.281	8.1
	28	1.413	8.8	<u>1.463</u>	<u>9.1</u>	2.808	17.8	1.494	9.3	1.475	9.2
	120	1.547	9.6	<u>1.547</u>	<u>9.6</u>	2.810	17.8	1.639	10.1	1.621	10.0
WEA-LD	4	1.723	15.2	1.814	15.8	4.668	34.9	<u>1.740</u>	<u>15.3</u>	1.811	16.0
	12	2.959	23.9	2.986	24.1	4.661	34.9	3.008	24.1	2.973	<u>24.0</u>
	28	3.216	25.5	3.271	25.8	4.659	34.9	3.378	26.5	<u>3.268</u>	<u>25.8</u>
	120	3.246	25.8	<u>3.319</u>	<u>26.2</u>	4.668	34.9	3.667	28.5	3.407	26.7
WEA-NY	4	<u>1.272</u>	11.9	1.331	12.4	5.386	40.9	1.268	11.8	1.366	12.9
	12	<u>2.448</u>	21.2	2.474	21.6	5.384	40.9	2.510	21.8	2.438	<u>21.5</u>
	28	2.674	23.1	<u>2.708</u>	23.4	5.387	40.9	2.797	23.8	2.724	<u>23.1</u>
	120	2.713	23.2	<u>2.788</u>	<u>23.5</u>	5.389	40.9	3.063	25.6	2.911	24.6
WEA-SG	4	0.344	1.3	0.346	1.3	0.700	2.5	0.341	1.2	0.346	1.3
	12	0.416	1.5	<u>0.421</u>	<u>1.5</u>	0.698	2.5	0.424	1.5	0.425	1.5
	28	0.465	1.7	0.473	1.7	0.698	2.5	0.495	1.8	<u>0.471</u>	<u>1.7</u>
	120	0.483	1.8	0.496	1.8	0.697	2.5	0.512	1.9	<u>0.485</u>	<u>1.8</u>

Table 3: Performance of Sonnet across the ILI and WEA tasks with different attention modules. Best results are **bolded**, second best underlined. MAE values are rounded to 3 decimals, sMAPE to 1 for spacing.

et al. 2022), the Frequency Enhanced Decomposed (FED) attention proposed in FEDformer (Zhou et al. 2022a), the variable deformable attention (VDAB) proposed by DeformTime (Shu and Lamos 2025), and ours (MVCA). Further details are provided in Appendices C and D.

Table 2 enumerates MAEs averaged across the 4 test seasons for all forecasting horizons (sMAPEs in Appendix E, Table S9). Notably, for the base models, removing the attention module does not lead to significant performance degradation, with some cases even showing improved results, e.g. improved accuracy is reported for some horizons for the ILI-ENG task. This indicates that the attention modules in the base models do not always capture useful information.

By replacing naïve attention with the proposed MVCA module, we obtain significant performance gains across all ILI tasks, reducing MAE by 10.7% on average across all base models. The average MAE reduction for PatchTST is greater (15.1%) compared to the other base models (10.4% for Samformer and 6.7% for iTransformer), indicating that the application of MVCA yields enhanced performance gains

for the model that does not capture inter-variable dependencies. When compared to other modified attention modules, MVCA shows competitive performance, reducing MAE by 2.8% on average. Compared to the best-performing baseline (VDAB from DeformTime), it reduces MAE and sMAPE by 3.5% and 3%, respectively. While the gains in the ENG region are less pronounced, MVCA achieves consistently the best results across the US regions, and VDAB is the second best; both methods capture inter-variable dependencies while others do not. Comparably, replacing naïve attention with the Fourier attention modules from FEDformer and FNet results in worse performance.

4.4 Sonnet with Different Attention Variants

To understand how different attention mechanisms affect the forecasting accuracy of Sonnet, we replace the MVCA module with the aforementioned modified attention mechanisms (see section 4.3). Experiments are conducted on both the ILI and WEA forecasting tasks across all locations, forecasting horizons, and test periods. Hyperparameters are re-tuned for each attention variant using the same validation procedure. Appendix C provides further details.

Results are enumerated in Table 3. Overall, using MVCA yields the best accuracy compared to other attention modules, outperforming them in 26 out of the 32 tasks (MAE). On average, it reduces the MAE and sMAPE scores by 2.9% and 2%, respectively. The most competitive baseline module is VDAB (DeformTime). MVCA outperforms it by 3.6% and 3% on average w.r.t. MAE and sMAPE, respectively. The worst-performing attention module is FED (FEDformer). MVCA reduces its MAE by 51.5%. FED focuses on capturing seasonality information using frequency transformation without considering the dependencies between variables. This shows that modelling seasonality alone is insufficient for MTS forecasting, particularly when key predictive information comes from exogenous variables. The performance obtained by using either naïve attention or FNet is similar. MVCA reduces their MAE by 7% and 6.6% on average, respectively. This is expected as FNet does not introduce learnable parameters to the naïve attention.

5 Conclusion

In this paper, we present Sonnet, a novel model for multivariable time series forecasting tasks. Sonnet operates in the spectral dimension using learnable wavelet transforms and captures variable dependencies with frequency-based coherence. It then predicts future temporal dynamics with the Koopman operator. Experiments are conducted on 12 data sets from different application domains, with 9 of them containing multiple test seasons for a more comprehensive analysis. Sonnet yields the best performance in 34 out of 47 tasks, reducing MAE by 2.2% and 1.1% on average compared to the most performant baseline overall (DeformTime) or per task (can vary), respectively. Additionally, our experiments highlight that replacing vanilla attention, present in various forecasting models, with the proposed MVCA module improves prediction accuracy by a large margin.

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